HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and optimized **PyTorch** ML compilers for a specialized **NPU architecture**, focusing on efficient graph transformations and operator fusion.
- Implemented advanced **quantization techniques** in **C++** to reduce model size and inference latency for neural network deployments on embedded hardware.
- Analyzed and debugged **ML workloads** on target hardware, identifying performance bottlenecks in complex computational graphs and resolving them through code optimization.
- Collaborated with architecture teams to integrate new ML compiler features and system-level optimizations into the core platform, improving overall performance by approximately **15%**.
- Researched and prototyped novel solutions for hardware-aware ML optimization, contributing to the design of next-generation NPU architectures.
- Created detailed technical documentation and internal training materials on compiler design and quantization methodologies for team knowledge sharing.
- Containerized ML compiler toolchains and deployment pipelines using **Docker** and managed them within a **Kubernetes** cluster for consistent development and testing environments.

SKILLS

ML Development – PyTorch, C++, ML Workloads, Quantization Techniques, Compiler Development, Performance Optimization, ML Model Deployment

Programming Languages – Python, C++, Java

Cloud & DevOps – Docker, Kubernetes, AWS (EC2, S3, Lambda, CloudWatch), Jenkins, Git

Data & Analytics – SQL, pandas, NumPy, scikit-learn, Power BI, Tableau, R / RStudio

Enterprise Systems – Spring Boot, REST APIs, ERP Integration Support, Change Management

Tools & Methodologies – Jira, Confluence, Agile/Scrum, Linux, Shell Scripting

PROJECTS

ML Compiler Optimization for NPU Architectures

- Designed and implemented a **PyTorch-based compiler pass** to optimize tensor operations for a custom NPU, achieving a **22% reduction** in execution cycles for key ML models.
- Developed **C++** modules for parameter quantization, enabling models to run with **8-bit integers** while maintaining a less than **1% drop** in accuracy for convolutional neural networks.
- Integrated the compiler toolchain into a **CI/CD pipeline** using **Jenkins**, automating build, test, and deployment processes for new compiler versions.

Hardware-Accelerated ML Inference Engine

- Engineered a **FastAPI** service in **Python** to serve ML models optimized for NPU deployment, handling approximately **150 requests per second**.
- Implemented low-level **C++** inference kernels, directly interacting with hardware acceleration libraries to minimize latency for real-time ML tasks.
- Created a Docker image for the inference engine, ensuring consistent deployment across development and staging environments managed via **Kubernetes**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)